

Serverless Layer — Monthly Cost Estimate (af-south-1)

Assumptions: Steady State

Metric	Value
Orders per day	500
Orders per month	15,000
Average Lambda 1 duration	230ms
Average Lambda 2 duration	150ms
Average payload size	1KB
Lambda memory	128MB

Assumptions: 50x Spike (peak month)

Metric	Value
Orders per day (spike days x5)	25,000
Orders per month	75,000
Everything else same	

API Gateway: HTTP API

	Steady State	50x Peak
Requests	15,000	75,000
Price per million requests	\$1.00	\$1.00
Monthly cost	~\$0.02	~\$0.08

Lambda 1: Order Processor

Item	Steady State	50x Peak
Invocations	15,000	75,000
Duration per invoke	230ms	230ms
Memory	128MB (0.125GB)	
GB-seconds	431	2,156
Free tier (GB-seconds)	400,000/month	400,000/month
Invocation free tier	1,000,000/month	1,000,000/month
Monthly cost	\$0.00	\$0.00

Lambda 2: Notification Processor

Item	Steady State	50x Peak
Invocations	15,000	75,000
Duration per invoke	150ms	150ms
Memory	128MB (0.125GB)	
GB-seconds	281	1,406
Free tier (GB-seconds)	400,000/month	400,000/month
Monthly cost	\$0.00	\$0.00

SQS: OrderConfirmations Queue

Item	Steady State	50x Peak
Messages sent (Lambda 1 → SQS)	15,000	75,000
Messages received (SQS → Lambda 2)	15,000	75,000
Total requests	30,000	150,000
Free tier	1,000,000 requests/month	1,000,000 requests/month
Monthly cost	\$0.00	\$0.00

SNS — LogiHaul-OrderAlerts

Item	Steady State	50x Peak
Notifications published	15,000	75,000
Email deliveries	15,000	75,000
Free tier (publishes)	1,000,000/month	1,000,000/month
Email delivery cost	\$2.00 per 100,000	\$2.00 per 100,000
Monthly cost	~\$0.30	~\$1.50

DynamoDB — LogiHaul-Orders (On-Demand)

Item	Steady State	50x Peak
Write requests (orders)	15,000	75,000
Read requests (GSI queries)	~30,000	~150,000
Storage (est. 1KB per item)	~15MB	~75MB
Write free tier	1,000,000 WRU/month	1,000,000 WRU/month
Read free tier	25 RCU provisioned equivalent	—
Monthly cost	\$0.00	\$0.00

Serverless Layer — Total Summary

Service	Steady State	50x Peak
API Gateway (HTTP API)	\$0.02	\$0.08
Lambda 1	\$0.00	\$0.00
Lambda 2	\$0.00	\$0.00
SQS	\$0.00	\$0.00
SNS (email delivery)	\$0.30	\$1.50
DynamoDB (on-demand)	\$0.00	\$0.00
TOTAL	~\$0.32/month	~\$1.58/month

The serverless layer costs approximately \$0.32/month at steady state and \$1.58/month during a 50x spike, both figures well within LogiHaul's \$150/month steady-state and \$400/month peak budget.

The dominant cost is SNS email delivery at \$2.00 per 100,000 notifications. Lambda, SQS, and DynamoDB on-demand all remain within their permanent free tier limits at LogiHaul's current order volumes, meaning the serverless intake pipeline scales from zero to 50x traffic with negligible incremental cost, the core financial justification for choosing serverless over a provisioned alternative.